



Post Office Box 2000, Decatur, Alabama 35609-2000

August 4, 2023

10 CFR 50.73
10 CFR 50.4(a)

ATTN: Document Control Desk
U.S. Nuclear Regulatory Commission
Washington, D.C. 20555-0001

Browns Ferry Nuclear Plant, Unit 2
Renewed Facility Operating License No. DPR-52
NRC Docket No. 50-260

Subject: **Licensee Event Report 50-260/2023-002-00 – Main Steam Relief Valves Lift Settings Outside of Technical Specifications Required Setpoints**

The enclosed Licensee Event Report provides details of Main Steam Relief Valves which failed to meet their Surveillance Requirements for longer than allowed by plant Technical Specifications. The Tennessee Valley Authority is submitting this report in accordance with Title 10 of the Code of Federal Regulations 50.73(a)(2)(i)(B), as any operation or condition which was prohibited by the plant's Technical Specifications.

There are no new regulatory commitments contained in this letter. Should you have any questions concerning this submittal, please contact Michael W. Oliver, Acting Site Licensing Manager, at (256) 729-2636.

Respectfully,

A handwritten signature in black ink, appearing to read "Joe Quinn", is written over a horizontal line.

Joe Quinn by delegation for SVP
Manu Sivaraman
Site Vice President

Enclosure: Licensee Event Report 50-260/2023-002-00 – Main Steam Relief Valves Lift Settings Outside of Technical Specifications Required Setpoints

U.S. Nuclear Regulatory Commission

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cc (w/ Enclosure):

NRC Regional Administrator - Region II

NRC Senior Resident Inspector - Browns Ferry Nuclear Plant

NRC Project Manager - Browns Ferry Nuclear Plant



LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form
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1. Facility Name

Browns Ferry Nuclear Plant, Unit 2

☒ 050
☐ 052

2. Docket Number

000260

3. Page

1 OF 7

4. Title

Main Steam Relief Valves Lift Settings Outside of Technical Specifications Required Setpoints

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved		
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name	☐ 050	Docket Number
06	05	2023	2023	- 002	- 00	08	04	2023	N/A	☐ 050	N/A
									N/A	☐ 052	N/A

9. Operating Mode

1

10. Power Level

100

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)

10 CFR Part 20	☐ 20.2203(a)(2)(vi)	10 CFR Part 50	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.1200(a)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	☐ 73.1200(b)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	☐ 73.1200(c)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)	☐ 73.1200(d)
☐ 20.2203(a)(2)(i)	10 CFR Part 21	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	10 CFR Part 73	☐ 73.1200(e)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.77(a)(1)	☐ 73.1200(f)
☐ 20.2203(a)(2)(iii)		☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(2)(i)	☐ 73.1200(g)
☐ 20.2203(a)(2)(iv)		☒ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(ii)	☐ 73.1200(h)
☐ 20.2203(a)(2)(v)		☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)		
☐ OTHER (Specify here, in abstract, or NRC 366A).					

12. Licensee Contact for this LER

Licensee Contact

Ryan Coons, Licensing Engineer

Phone Number (Include area code)

256-729-2070

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS
B	SB	RV	T020	N	N/A	N/A	N/A	N/A	N/A

14. Supplemental Report Expected

☒ No ☐ Yes (If yes, complete 15. Expected Submission Date)

15. Expected Submission Date

Month	Day	Year

16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)

On June 5, 2023, the Tennessee Valley Authority was notified of as-found testing results that four (4) Main Steam Relief Valves (MSRVs) from Browns Ferry Nuclear Plant (BFN), Unit 2 were outside of their Surveillance Requirement limits for an indeterminate amount of time. These affected valves remained capable of maintaining reactor pressure within the American Society of Mechanical Engineers code limits and were still able to perform their safety function.

It was determined that one MSRV failed due to corrosion bonding to the valve seats and non-quantifiable leakage known as simmering. Three MSRVs failed below their setpoints due to relaxed setpoint springs. All thirteen of the MSRV pilot valves were replaced during the Unit 2 Spring 2023 refueling outage. The Boiling Water Reactor Owners' Group is continuing to work toward a solution to improve the quality and adhesion of the platinum coating on the discs. BFN will take additional actions to verify that the springs associated with each low failure passes Spring Rate testing or is replaced.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

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1. FACILITY NAME Browns Ferry Nuclear Plant, Unit 2	☒ 050	2. DOCKET NUMBER 00260	3. LER NUMBER		
	☐ 052		YEAR 2023	SEQUENTIAL NUMBER - 002	REV NO. - 00

NARRATIVE**I. Plant Operating Conditions before the Event**

At the time of discovery, Browns Ferry Nuclear Plant (BFN) Unit 2 was in Mode 1 at approximately 100 percent power.

II. Description of Event**A. Event Summary**

On June 5, 2023, NTS notified the Tennessee Valley Authority (TVA) with the as-found testing results of the thirteen Main Steam Relief Valves (MSRVs) [RV], which were removed during the Spring 2023 Unit 2 Refueling Outage 22 (U2R22). Four (4) MSRVs (BFN-2-PCV-001-0004, BFN-2-PCV-001-0019, BFN-2-PCV-001-0031, and BFN-2-PCV-001-0179) had as-found lift settings which were outside of the +/- 3 percent band of their setpoints required by Technical Specification (TS) 3.4.3, Safety/Relief Valves (S/RVs) for an indeterminate amount of time.

Throughout this event, the two stage MSRV pilot valves remained capable of maintaining reactor pressure below 1375 pounds per square inch gauge (psig), which is the American Society of Mechanical Engineers (ASME) code limit of 110 percent of the vessel design pressure. The valves remained capable of performing their required safety function.

The TVA is submitting this report in accordance with Title 10 of the Code of Federal Regulations 50.73(a)(2)(i)(B), as any operation or condition which was prohibited by the plant's TS.

B. Status of structures, components, or systems that were inoperable at the start of the event and that contributed to the event

There were no structures, systems, or components (SSCs) whose inoperability contributed to this event.

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NARRATIVE**C. Dates and approximate times of occurrences**

Dates and Approximate Times	Occurrence
April 21, 2021	Unit 2 entered Mode 2, beginning Fuel Cycle 22 (U2C22).
February 17, 2023	Unit 2 entered Mode 4 for U2R22.
June 5, 2023	NTS notified the TVA with as-found testing results of the thirteen Unit 2 MSRV pilot valves removed during U2R22.

D. Manufacturer and model number of each component that failed during the event

The failed components were all Target Rock Corporation two stage pressure control valves, model number 7567F.

E. Other systems or secondary functions affected

No other systems or secondary functions were affected.

F. Method of discovery of each component or system failure or procedural error

The MSRV failures were discovered at NTS during as-found testing of the thirteen MSRV two stage pilot valves which were removed during U2R22.

G. The failure mode, mechanism, and effect of each failed component

MSRV BFN-2-PCV-001-0004 failed due to corrosion bonding to the valve seat and a non-quantifiable leakage known as simmering.

MSRVs BFN-2-PCV-001-0019, BFN-2-PCV-001-0031, and BFN-2-PCV-001-00179 failed due to a relaxation of the setpoint spring over time.

H. Operator actions

There were no operator actions associated with this event.

I. Automatically and manually initiated safety system responses

There were no automatic or manual safety system responses associated with this event.

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CONTINUATION SHEET**

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NARRATIVE**III. Cause of the event****A. Cause of each component or system failure or personnel error**

MSRV 2-PCV-001-0004 failed due to corrosion bonding to the valve seat and a non-quantifiable leakage known as simmering. As a result of corrosion bonding, the force required to break the crystal structure of the corrosion bond alters the mechanical setpoint of the pilot valve. This issue is commonly known to the industry as set point drift. Simmering valves result in mechanical setpoint drift and is generally the result of low stellite in the seat.

MSRVs BFN-2-PCV-001-0019, BFN-2-PCV-001-0031, and BFN-2-PCV-001-00179 failed below their setpoints due to relaxed setpoint springs.

B. Cause(s) and circumstances for each human performance related root cause

No human performance related root causes were identified.

IV. Analysis of the event

BFN, Unit 2 TS Surveillance Requirement (SR) 3.4.3.1 specifies verification of lift settings of the required 12 Safety/Relief Valves (S/RVs) are within a +/- 3 percent band of their setpoint values in accordance with the Inservice Testing Program. BFN, Unit 2 has thirteen MSRVs to satisfy this requirement with margin. When tested, the following S/RVs were outside the allowable +/- 3 percent band:

<u>S/RV ID Number</u>	<u>Setpoint (psig)</u>	<u>Test Result (psig)</u>	<u>Difference (Percent)</u>
BFN-2-PCV-001-0004	1155	1217	+ 5.37
BFN-2-PCV-001-0019	1135	1070	- 5.73
BFN-2-PCV-001-0031	1135	1099	- 3.17
BFN-2-PCV-001-0179	1155	1114	- 3.55

Prior to startup from U2R22 all thirteen MSRV pilot valves were replaced with refurbished valves which were certified to lift within +/- 1 percent of their setpoint. Operating Experience has shown that Target Rock two stage MSRV setpoint drift is not a uniform, linear process. The corrosion bonding increases at a random rate.

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NARRATIVE**V. Assessment of Safety Consequences**

System availability was not impacted by this event. The failure of the MSRV pilot valves to meet their TS 3.4.3 specified mechanical setpoints did not impact their remote manual operation or activation through the MSRV Automatic Actuation Logic, since these operating modes and functions rely upon electrically signaled control air solenoids to open the MSRV pilot valves.

TS Bases 3.4.3 states that the overpressure protection system must accommodate the most severe pressurization transient. The MSRVs remained capable of maintaining the reactor pressure below 1375 psig, which is the ASME code limit (110 percent of the vessel design pressure). The valves remained capable of performing their required safety function.

Based on the above, the TVA has concluded that sufficient systems were available to provide the required safety functions needed to protect the health and safety of the public.

A. Availability of systems or components that could have performed the same function as the components and systems that failed during the event

Each BFN operating unit has a non-safety related, electrical logic system (MSRV Actuation Logic) installed, which provides defense in depth against MSRV setpoint drift by electrically opening MSRV groups based upon setpoints at 1135 psig, 1145 psig, and 1155 psig. Therefore, during a reactor pressure transient event, the four 1135 psig group MSRVs, followed by the four 1145 psig group MSRVs, and finally the five 1155 psig group MSRVs would receive an electrical open signal, providing a defense in depth function to allow the valves to perform their safety function.

B. For events that occurred when the reactor was shut down, availability of systems or components needed to shutdown the reactor and maintain safe shutdown conditions, remove residual heat, control the release of radioactive material, or mitigate the consequences of an accident

This event did not occur when the reactor was shutdown.

C. For failure that rendered a train of a safety system inoperable, estimate of the elapsed time from discovery of the failure until the train was returned to service

TS 3.4.3 requires twelve of the thirteen S/RVs to be operable for S/RV system operability. While four of thirteen valves failed to meet their TS SR 3.4.3.1 limits for an indeterminate period, they remained capable of performing their required safety function.

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NARRATIVE**VI. Corrective Actions**

Corrective Actions are being managed by the TVA's corrective action program under Condition Reports (CRs), 962223, 1286467, 1410577, 1521190, 1658693, 1699286, 1775232, 1822254, and 1860559.

A. Immediate Corrective Actions

All thirteen of the BFN, Unit 2 MSRV pilot valves were replaced with refurbished valves during U2R22. As left testing verified that these refurbished pilot valves were within +/- 1 percent of their setpoints.

B. Corrective Actions to Prevent Recurrence or to reduce the probability of similar events occurring in the future

As discussed in LER 50-260/2021-002-00, a flaking issue has been noted with the platinum coated pilot discs. The Boiling Water Reactor Owners' Group (BWROG) is continuing to work toward a solution to improve the quality and adhesion of the platinum coating on the discs. The corrective actions suggested by the BWROG will be incorporated to correct setpoint drift. To reduce the probability of seat leakage occurring in the future, pilot seat rebuilds will be performed in valves that have low stellite in the seat.

BFN will take additional actions to verify that the springs associated with each low failure passes Spring Rate testing or is replaced.

VII. Previous Similar Events at the Same Site

A search of LERs from BFN, Units 1, 2, and 3 over the last five years identified six LERs associated with MSRV lift settings outside of TS required setpoints:

- LER 50-259/2022-003-00, for Unit 1 Cycle 14
- LER 50-296/2022-001-00, for Unit 3 Cycle 20
- LER 50-260/2021-002-00, for Unit 2 Cycle 21
- LER 50-259/2020-003-01, for Unit 1 Cycle 13
- LER 50-296/2020-002-00, for Unit 3 Cycle 19
- LER 50-260/2019-002-00, for Unit 2 Cycle 20

VIII. Additional Information

There is no additional information.



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NARRATIVE

IX. Commitments

There are no new commitments.